AN ETALE NEIGHBOURHOOD IS DOMINATED AFTER A FINITE LOCALLY FREE BASE CHANGE OVER A ZARISKI NEIGHBOURHOOD

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 11822-11838 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 11840-11849, the complete author proof (8 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

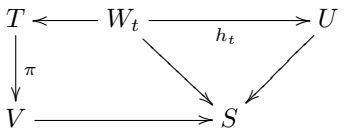
Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Every etale morphism is locally standard etale (Morphisms), and the algebra lemma that standard etale neighbourhoods of affine schemes are dominated by finite free rings after Zariski localization; cited inputs with the reduction written in the retained span.

Source reference. Original lemma 00UF: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11849–11849. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02GT: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11849–11849. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $s \in S$. Let $f : (U, u) \to (S, s)$ be an étale neighbourhood. There exists an affine open neighbourhood $s \in V \subset S$ and a surjective, finite locally free morphism $\pi : T \to V$ such that for every $t \in \pi^{-1}(s)$ there exists an open neighbourhood $t \in W_t \subset T$ and a commutative diagram*



with $h_t(t) = u$.

Proof. The problem is local on S hence we may replace S by any open neighbourhood of s. We may also replace U by an open neighbourhood of u. Hence, by

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Morphisms, Lemma 02GT we may assume that $U \to S$ is a standard étale morphism of affine schemes. In this case the lemma (with $V = S$) follows from Algebra, Lemma 00UF. $\square$